Aproximare în sensul CMMP

↳ celor mai mici pătrate

matematician ↗

Spații pre-Hilbertiene (F, u)

$u: F \times F \to \mathbb{R}$ $\quad f \in F$ $\quad u = \langle \cdot, \cdot \rangle$

↳ produs scalar cu proprietățile

1. **Liniaritate**

$$\begin{cases} \langle f_1 + f_2, f_3 \rangle = \langle f_1, f_3 \rangle + \langle f_2, f_3 \rangle \\ \langle \alpha f_1, f_2 \rangle = \alpha \langle f_1, f_2 \rangle \end{cases}$$

2. **Definirea pozitivă**

$$\langle f, f \rangle \geq 0$$

3. **Nesingularitate**

$$\langle f, f \rangle = 0 \iff f = 0_F$$

Exemple → (produs scalar)

$F = \mathbb{R}^3 \to$ $\langle x, y \rangle = \sum_{i=1}^{3} x_i y_i$

$F = C([a,b]) \Rightarrow$ $\langle f, g \rangle \int_a f(x) \cdot g(x) \, \omega(x) \, dx$

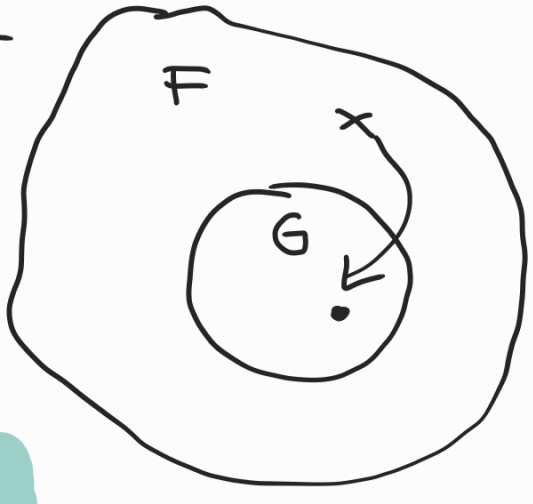
ponderare ↑

$$\langle f, g \rangle = \sum_{i=1}^{n} f(x_i) \cdot g(x_i) \cdot \omega(x_i)$$

c) $F = \mathbb{R}[x]$

- $G \subseteq F$ subspațiu finit cu elemente liniar independente
- Norma unui element : $\|f\| = \sqrt{\langle f, f \rangle}$

Cel mai bun aproximant în sensul CMMP pt. un element $f \in F$ este un element $g^* \in G$ cu proprietatea



$$\|f - g^*\| = \min_{g \in G} \|f - g\|$$

Teorema 1

Condiția necesară și suficientă ca $g^* \in G \subset F$ să fie cel mai bun aproximant în sensul CMMP pentru $f \in F$ este

$$\langle f - g^*, g \rangle = 0 \quad (\forall) g \in G$$

Demonstrații matematice

a) **Condiție necesară**

Presupunem că există $g_1 \in G$ a.î.

$$\langle f - g^* \mid g_1 \rangle = k, \quad k \neq 0$$

$$g_2 = g^* + \frac{k}{\|g_1\|^2} \cdot g_1 \quad (\text{ales din } G)$$

$$\|f - g_2\|^2 = \langle f - g_2, f - g_2 \rangle = \langle f - g^* - \frac{k}{\|g_1\|^2} \cdot g_1,$$

$$f - g^* - \frac{k}{\|g_1\|^2} \cdot g_1 \rangle = \langle f - g^*, f - g^* \rangle -$$

$$2\frac{k}{\|g_1\|^2} \underbrace{\langle f - g^*, g_1 \rangle}_{k} + \frac{k^2}{\|g_1\|^4} \underbrace{\langle g_1, g_1 \rangle}_{\|g_1\|^2}$$

$$= \|f - g^*\|^2 - \frac{k^2}{\|g_1\|^2} \rightarrow$$

$$\|f - g_2\| < \|f - g^*\| \quad \text{Fals!} \Rightarrow \boxed{k = 0}$$

b) **Condiție suficientă**

Fie un $g_1 \in G$ a.î. $\langle f - g_1, g \rangle = 0 \ (\forall) g \in G$.

$$\|f - g\|^2 = \langle f - g_1, f - g \rangle = \langle f - g_1 - (g - g_1),$$

$$f - g_1 - (g - g_1) \rangle = \|f - g_1\|^2 + \|g - g_1\|^2$$

$$\Rightarrow \|f - g_1\|^2 < \|f - g\|^2 \ (\forall) g \in G \Rightarrow \boxed{g_1 = g^*}$$

Teorema 2 $\rightarrow g^*$ este unic!

Dem: Există g_1^* și g_2^* c.m.b. a s.c.m.m.p. definiți

$$\langle f - g_1^* | g \rangle = \langle f - g_2^* | g \rangle = 0 \ (\forall) \ g \in G$$

$g = g_1^* - g_2^*$ (alegem)

$$\| g_1^* - g_2^* \|^2 = \langle g_1^* - f + f - g_2^* | g \rangle$$

$$= \langle f - g_2^* | g \rangle - \langle f - g_1^* | g \rangle = 0$$

$$\Rightarrow \| g_1^* - g_2^* \| = 0 \Rightarrow g_1^* = g_2^*$$

Construirea lui g^* $\rightarrow$ finită

Fie $u_1, u_2, \dots, u_n$ o bază pt. $G \Rightarrow$

$$g \in G \quad g = \sum_{i=1}^{N} c_i \cdot u_i$$

$$g^* \in G, \ g^* = \sum_{i=1}^{N} c_i^* \cdot u_i$$

$$\langle f - g^* | g \rangle = \langle f - g^* | \sum_{i=1}^{n} c_i u_i \rangle =$$

$$= \sum_{i=1}^{n} c_i \cdot \langle f - g^*, u_i \rangle = 0 \Rightarrow$$

$$\boxed{\langle f - g^* | u_i \rangle = 0, \ i = \overline{1:n}}$$

$$\langle f, u_i \rangle - \langle g^*, u_i \rangle = 0 \Rightarrow$$

$$\langle f, u_i \rangle = \langle g^*, u_i \rangle, \quad i = 1:n$$

$$\left\langle \sum_{k=1}^{n} c_k^* u_k, u_i \right\rangle = \langle f, u_i \rangle$$

$$\sum_{k=1}^{n} c_k^* \langle u_k, u_i \rangle = \langle f, u_i \rangle, \quad i = 1:n$$

$$\begin{cases} c_1^* \langle u_1, u_1 \rangle + c_2^* \langle u_2, u_1 \rangle + \dots + c_n^* \langle u_n, u_1 \rangle = \langle f, u_1 \rangle \\ c_1^* \langle u_1, u_2 \rangle + c_2^* \langle u_2, u_1 \rangle + \dots + c_n^* \langle u_n, u_2 \rangle = \langle f, u_2 \rangle \\ \vdots \\ c_1^* \langle u_1, u_n \rangle + c_2^* \langle u_2, u_n \rangle + \dots + c_n^* \langle u_n, u_n \rangle = \langle f, u_n \rangle \end{cases}$$

(SEL)

$u_1, u_2, \dots, u_n \rightarrow$ bază ortogonală

$\langle u_i, u_j \rangle = 0, \ i \neq j \Rightarrow$ sistem diagonală

ortonormată $\langle u_i, u_i \rangle = 1$

$$c_k^* = \langle f, u_k \rangle \quad k = 1:n$$ (coeficienți Fourier)

$\hookrightarrow O(N)$

Exemplu 1 → Aproximare trigonometrică

bază: $\frac{1}{\sqrt{2}}$ $\sin x$ $\cos x$ $\sin 2x$ $\cos 2x$ ... $\sin nx$ $\cos nx$

$$\frac{1}{\pi}\int_0^{2\pi} u_q(x)\cdot u_r(x)\cdot \underbrace{1}_{w(x)}\,dx =$$

$$= \begin{cases} 0, & q \neq r \\ 1, & q = r \end{cases}$$

$f \to$ semnal continuu

$$g^*(x) = \frac{a_0}{\sqrt{2}} + \sum_{k=1}^{n}\left[a_k\cos(kx) + b_k\sin(kx)\right]$$

$$\begin{cases} a_0 = \langle f, u_0\rangle = \frac{1}{\pi\sqrt{2}}\int_0^{2\pi} f(x)\,dx \\ a_k = \langle f, u_k\rangle = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos kx\,dx \\ b_k = \langle f, u_k\rangle = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin kx\,dx \end{cases}$$

Exemplu 2 → aproximare folosind polinoame Cebâșev

$$T_n(x) = \cos(n \cdot \arccos x)$$

$T_0(x) = 1$

$T_1(x) = x$

$T_2(x) = 2x^2 - 1$

$T_3(x) = 4x^3 - 3x$

$T_n : \mathbb{R} \to [-1, 1]$

Rădăcini

$T_n(x_k) = 0$

$\cos(n \arccos x_k) = 0$

$n \arccos x_k = \frac{\pi}{2} + k\pi = \frac{2k+1}{2} \cdot \pi$

$$x_k = \cos\left(\frac{2k+1}{2n} \cdot \pi\right), \; k = 0 : n-1 \quad \in [-1, 1]$$

Extreme

$T_n(x_p) = \pm 1$

$\cos(n \arccos x_p) = \pm 1$

$n \arccos x_p = p\pi$

$$x_p = \cos\frac{p\pi}{n}, \; p = \overline{0, n-1}$$

Recurență

$$\begin{cases} T_0(x) = 1 \quad T_1(x) = x \\ T_{n+1}(x) - 2x\,T_n(x) + T_{n-1}(x) = 0 \end{cases}$$

Produs scalar

$$\int_{-1}^{1} T_p(x) \cdot T_q(x) \cdot \underbrace{\frac{1}{\sqrt{1-x^2}}}_{\omega(x)} dx = \begin{cases} 0, & p \neq q \\ \pi, & p = q = 0 \\ \frac{\pi}{2}, & p = q \neq 2 \end{cases}$$

$\omega(x)$ funcție pondere

$$P^*(x) = \frac{a_0 \cdot T_0(x)}{\sqrt{2}} + \sum_{i=1}^{n} a_i T_i(x)$$

Bază: $\frac{T_0}{\sqrt{2}}, T_1, T_2, \dots, T_n$

$$\begin{cases} a_0 = \frac{1}{\pi\sqrt{2}} \int_{-1}^{1} \frac{f(x)}{\sqrt{1-x^2}} dx \\ a_i = \frac{2}{\pi} \int_{-1}^{1} \frac{f(x) \cdot T_i(x)}{\sqrt{1-x^2}} dx \end{cases}$$

Aproximare discretă $f \to (x_i, f_i)$

1) trigonometric

$$\sum_{x=1}^{2n+2} u_p(x_p) \cdot u_q(x_k) = \begin{cases} 0, & p \neq q \\ n+1, & p = q \end{cases}$$

$x_k =$ echidistante δn $[0, 2\pi]$

$$x_k = \frac{2\pi k}{2n+2} = \frac{k\pi}{n+1} \qquad k = 1 : 2n+2 \; (0 : 2n+1)$$

$$a_0 = \langle f, \frac{1}{\sqrt{2}} \rangle = \frac{1}{\sqrt{2}} \sum_{k=1}^{2n+2} f(x_k)$$

$$a_p = \langle f, \cos px \rangle = \sum_{k=1}^{2n+2} f(x_k) \cdot \cos(p x_k)$$

$$b_p = \langle f, \sin px \rangle = \sum_{k=1}^{2n+2} f(x_k) \cdot \sin(p \cdot x_k)$$

2) Cebâșev

suportul $T_{n+1}(x_k) = 0 \Rightarrow x_k = \cos \frac{(2k+1)\pi}{2n+2}$

$$\sum_{k=0}^{n} T_p(x_k) \cdot T_q(x_k) = \begin{cases} 0, & p \neq q \\ \frac{n+1}{2}, & p = q \end{cases}$$

$$a_0 = \frac{\sqrt{2}}{n+1} \cdot \sum_{k=0}^{n} f(x_k)$$

$$a_i = \frac{2}{n+1} \cdot \sum_{k=0}^{n} f(x_k) \cdot T_i(x_k)$$